

LEGALBENCH: A COLLABORATIVELY BUILT BENCHMARK FOR MEASURING LEGAL REASONING IN LARGE LANGUAGE MODELS

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ABSTRACT

The advent of large language models (LLMs) and their adoption by the legal community has given rise to the question: what types of legal reasoning can LLMs perform? To enable greater study of this question, we present LEGALBENCH: a collaboratively constructed legal reasoning benchmark consisting of 162 tasks covering six different types of legal reasoning. LEGALBENCH was built through an interdisciplinary process, in which we collected tasks designed and hand-crafted by legal professionals. Because these subject matter experts took a leading role in construction, tasks either measure legal reasoning capabilities that are practically useful, or measure reasoning skills that lawyers find interesting. To enable cross-disciplinary conversations about LLMs in the law, we additionally show how popular legal frameworks for describing legal reasoning—which distinguish between its many forms—correspond to LEGALBENCH tasks, thus giving lawyers and LLM developers a common vocabulary. This paper describes LEGALBENCH, presents an empirical evaluation of 20 open-source and commercial LLMs, and illustrates the types of research explorations LEGALBENCH enables.

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1 Introduction

Advances in large language models (LLMs) are leading American lawyers and administrators to reexamine the practice of law [53, 63, 160, 57].² Proponents have argued that LLMs could alter how lawyers approach tasks ranging from brief writing to corporate compliance [160]. By making legal services more accessible, they could eventually help alleviate the United States’ long standing access-to-justice crisis [36, 134]. This perspective is informed by the observation that LLMs possess special properties which, it is argued, make them more suited for legal tasks. The models’ capacity to learn new tasks from limited labeled data would reduce the manual data annotation costs that ordinarily burden the development of legal language models [13]. Their apparent proficiency at sophisticated reasoning tasks would also make them ideal for the rigor of law, which requires parsing obtuse texts with heavy jargon, and inferential processes which combine different modalities of reasoning [157].

This excitement, however, is tempered by the fact that legal applications often involve significant risk [48]. Existing work has shown that LLMs are capable of generating content that is offensive, misleading, and factually incorrect [10, 80]. Such behaviors—if replicated in legal applications [114]—could result in substantial harms [146], with much of the potential burden imposed on traditionally marginalized and under-resourced populations [127, 138]. The safety implications thus create a pressing need to develop infrastructure and processes for benchmarking LLMs in legal contexts.

However, significant challenges face practitioners seeking to assess whether LLMs can perform legal reasoning. The first challenge is the limited ecosystem of legal benchmarks [157]. The majority of existing benchmarks, for example, focus on tasks which models learn by finetuning or training on task-specific data [21]. These benchmarks do not measure the aspects of LLMs which generate excitement for law—namely, their ability to perform many different tasks using only few-shot prompts. Relatedly, benchmarking efforts have focused on professional certification exams like the Uniform Bar Exam [71], but these are not always representative of the actual use-cases for LLMs. The second challenge is the incongruity between the ways in which existing benchmarks and lawyers frame “legal reasoning.” Existing benchmarks coarsely generalize all tasks involving legal data or laws as measuring “legal reasoning.” In contrast, lawyers recognize that legal reasoning is a broad umbrella term encompassing many distinct types of reasoning [47]. Different legal tasks require different skills and bodies of knowledge. Because existing legal benchmarks fail to draw these distinctions, it is difficult for legal professionals to contextualize the performance of modern LLMs within their own understanding of legal competency. In short: legal benchmarks do not use the same vocabulary or conceptual frameworks as the legal profession.

In light of these limitations, we believe that rigorously evaluating the legal reasoning capabilities of LLMs will require the legal community to take a more proactive role in the process of benchmarking. To that end, we present LEGALBENCH: the first steps towards constructing an interdisciplinary collaborative legal reasoning benchmark for the English language.³ Over the past year, the authors of this paper—drawing from their diverse legal and computer science backgrounds—came together to assemble 162 tasks (from 36 different data sources), each of which measures a specific type of legal reasoning. LEGALBENCH is thus, to the best of our knowledge, the first *open-source legal benchmarking effort*. We believe that this style of benchmark construction—where domain experts take an active and participatory role in the crafting of evaluation tasks—illustrates one approach to interdisciplinary collaboration in LLM research. Importantly, we believe it also shows that legal professionals have an essential role to play in the assessment and development of LLMs for law.

As a research project, we highlight three components of LEGALBENCH:

²In using “LLMs”, we are referring to language models which evince in-context learning capabilities (also referred to as “foundation models” [13]). This behavior has traditionally been observed in models with at least a billion parameters.

³<https://github.com/HazyResearch/legalbench/>

1. LEGALBENCH was constructed from a mix of existing legal datasets (restructured for the few-shot LLM paradigm), and hand-crafted datasets created and contributed by legal professionals (included as authors on this work). The legal professionals involved in this collaboration were asked to contribute datasets that they believed to either measure an interesting legal reasoning skill, or to capture a practically useful application for LLMs in the law. High performance on LEGALBENCH tasks thus provides useful information, allowing lawyers to validate their assessment of an LLM’s legal competency, or identify an LLM that could be used in their workflow.
2. LEGALBENCH tasks are organized into an extensive typology which describes the types of legal reasoning required to perform the task. Because this typology is drawn from frameworks familiar to the legal community, it enables legal professionals to meaningfully engage in discussions of LLM performance, using a terminology and conceptual framework familiar to them [47, 124].
3. Finally, LEGALBENCH is intended as a platform to support further research. For AI researchers who lack legal expertise, LEGALBENCH comes with significant support for understanding how to prompt and evaluate different tasks. And as more of the legal community begins to engage with the potential impact and role of LLMs, we hope to grow LEGALBENCH by continuing to solicit and incorporate tasks from legal professionals.⁴

In this paper, we make the following contributions:

1. First, we present a typology for organizing and describing legal tasks in terms of the types of reasoning they require. This typology is drawn from frameworks lawyers use to describe legal reasoning [124].
2. Second, we provide an overview of the tasks in LEGALBENCH, describing the process by which they were constructed, important dimensions of heterogeneity, and limitations. A full description of each task is provided in the Appendix.
3. Finally, we use LEGALBENCH to evaluate 20 LLMs from 11 different families, across a range of size points. We make observations regarding the performance of different models and present an initial study into different prompt-engineering strategies. Ultimately, these results are intended to highlight different directions of future work that LEGALBENCH may enable.

We hope that this benchmark will be interesting to a diverse set of communities. Practitioners may use these tasks to determine whether and where LLMs can be integrated into existing workflows to improve outcomes for clients. Legal academics may benefit from observing the types of annotation that LLMs are capable of [159], and different forms of empirical scholarly work they may enable. Computer scientists may benefit from studying the performance of these models in a domain like law, where distinct lexical properties and unique tasks may surface new insights.

Before we progress further, we note that the purpose of this work isn’t to evaluate whether computational systems *should* replace lawyers and legal officers, or to understand the positive and negative impacts of that replacement [48, 128, 4]. Rather, our goal is to construct artifacts that enable the relevant stakeholders and affected communities to better understand, *empirically*, the capacity for LLMs to perform different types of legal tasks. Given the proliferation of computational legal tools, we believe that answering this question is vital for ensuring their safe and ethical usage.

2 Related work

2.1 Legal reasoning benchmarks

Understanding the extent to which NLP models can perform tasks or skills traditionally associated with lawyers—or be useful in legal analysis—has been the focus of significant work [6, 110, 126, 84, 86, 104, 88, 22]. Researchers have approached this question in a variety of ways [72]. First, prior work has identified manually arduous tasks currently performed by lawyers—like forms of document review [62, 142] or case summarization [120, 121, 90, 69]—and developed benchmarks to assess the performance of current state-of-the-art techniques. Here, research has focused on the aspects of legal text which are often challenging for NLP methods, like the length of documents or the presence of jargon [21, 111, 87, 41, 79]. A second line of work has focused on developing tasks to evaluate forms of inferential reasoning common to law [21]. This includes, for instance, tasks which require a model to identify the best supporting statement for an argument [80, 157], or perform statutory reasoning [65]. Other work has focused on creating datasets for pretraining models [60, 129], non-English/multilingual tasks [95, 96, 66, 151, 52, 94, 70, 20, 23, 103, 18], legal judgement prediction [89, 38, 17, 158], legal role labeling [85], and different forms of retrieval [68].

⁴Cognizant of LEGALBENCH’s current skew towards American law, we hope that additional contributions incorporate tasks from other jurisdictions.

Importantly, the majority of previous benchmarking efforts have focused on language models which learn by supervised training or finetuning (e.g., BERT variants [45]), and researchers have consequently studied questions related to the role of domain specific datasets [157, 19, 20]. More recently, researchers have begun to ask whether *large* language models (LLMs) like GPT-3/4 can perform legal reasoning [73, 153, 67, 12, 29, 31, 154], citing to evidence of these models’ capacity to perform sophisticated reasoning tasks in domains like math or programming [145, 24]. Unlike BERT-based models, LLMs are evaluated on their ability to learn tasks *in-context*, primarily through prompting. While a few works have experimented with LLMs on existing benchmarks [16, 12], most evaluations focus on standardized tests or other exam equivalents [71, 93, 30]. Studies have explored the role of prompt-engineering [154, 153, 75], potential applications [93, 29, 147, 117, 116], questions regarding human-LLM interaction [31, 63], and comparisons to older finetuned-models [91].

LEGALBENCH builds on prior work in several ways. First, LEGALBENCH enhances opportunities to study legal reasoning in LLMs, by making available 162 evaluation tasks. LEGALBENCH systematizes and standardizes these tasks for LLM evaluation, specifying potential prompts, in-context demonstrations, and metrics. Second, LEGALBENCH presents a framework for organizing and comparing tasks, allowing researchers to identify trends in performance across groupings of tasks. This enables researchers, for instance, to distinguish between task types for which current LLMs are highly performant, and task types for which further work is needed.

A notable consequence of focusing on few-shot LLMs is that LEGALBENCH can contribute a much more diverse set of legal reasoning tasks. Traditional NLP methods require a large training set and a smaller evaluation set. The cost of legal annotations means that constructing benchmarks has required extraordinary financial investment [62, 120] or a “natural” source of existing annotations [157, 21]. Because the few-shot prompting regime requires only a few labeled demonstrations, creating large training sets isn’t necessary, and the effort they otherwise would have consumed can be allocated towards developing new tasks.

2.2 Connections to other LLM benchmarking efforts

We highlight connections to two broader research efforts. First, we draw inspiration from existing efforts within NLP and machine learning to define fine-grained measures of performance, which allow researchers to discuss model capabilities with precision and specificity. Examples include the diagnostic set of the GLUE Benchmark [141], the “reasoning patterns” studied in [100], the task organization used in HELM [80], and the BigBench effort [123]. Fine-grained measurements are valuable because they allow researchers to identify how particular modifications to model architectures or training regimes affect performance. They hold particular value for the field of legal NLP, in which researchers continue to debate how best to specialize language models to the domain [60, 157, 56]

We additionally draw inspiration from other large-scale collaborative efforts in AI, including the BigBench project [123], and studies in medicine [28]. In particular, we believe that LEGALBENCH illustrates a new model of open-source and interdisciplinary collaboration between the legal and AI communities. To the extent that LLMs gain adoption for legal tasks, legal professionals will be primarily charged with supervising them and selecting application use-cases. Involving the legal community in the design and construction of evaluation tasks allows for the construction of benchmarks which are more responsive to their interests and information needs.

3 The LEGALBENCH typology

LEGALBENCH identifies six types of legal reasoning that LLMs can be evaluated for: (1) issue-spotting, (2) rule-recall, (3) rule-application, (4) rule-conclusion, (5) interpretation, and (6) rhetorical-understanding. We first justify the selection of these types by providing background on how the legal profession frames “legal reasoning,” and the connections to our typology. We then illustrate how task datasets may be used to evaluate LLMs for each type, using examples from LEGALBENCH.

Though this framework draws heavily on American legal thought, we find it can be easily extended to characterize LEGALBENCH tasks that implicate non-American bodies of law. We also note that our types are non-exhaustive, and in future work hope to consider additions to these types.

3.1 Frameworks for legal reasoning

IRAC American legal scholars often describe “legal reasoning” as the process of determining the legal conditions that arise from a set of events or occurrences, with reference to both prior cases and codified laws [47]. A common framework for executing this type of legal reasoning is the **Issue, Rule, Application and Conclusion (IRAC)** framework [148, 124]. In this framework, legal reasoning decomposes into four sequential steps.

First, lawyers identify the legal issue in a given set of facts (**issue-spotting**). An issue is often either (1) a specific unanswered legal question posed by the facts, or (2) an area of law implicated in the facts. Depending on the setting, a lawyer may be told the issue, or be required to *infer* a possible issue.

Second, lawyers identify the relevant legal rules for this issue (**rule-recall**). A rule is a statement of law which dictates the conditions that are necessary (or sufficient) for some legal outcome to be achieved. In the United States, rules can come from a variety of sources: the Constitution, federal and state statutes, regulations, and court opinions (case law). Importantly, rules often differ between jurisdictions. Hence, the relevant rule in California might be different than the relevant rule in New York.

Third, lawyers apply these rules to the facts at hand (**rule-application**). Application, or the analysis of rule applicability, consists of identifying those facts which are most relevant to the rule, and determining how those facts influence the outcome under the rule. Application can also involve referencing prior cases involving similar rules (i.e. *precedent*), and using the similarities or differences to those cases to determine the outcome of the current dispute.

Finally, lawyers reach a conclusion with regards to their application of law to facts, and determine what the legal outcome of those facts are (**rule-conclusion**).

Example We illustrate this framework with a simple example. Suppose that BusinessMart—a large manufacturing corporation—is being sued by Amy in federal court on diversity jurisdiction.⁵ BusinessMart sells the majority of its goods in Texas, has its headquarters (where its CEO and board members sit and work) in California, and maintains a factory in Florida. A court is trying to determine—for the purposes of diversity jurisdiction—where BusinessMart’s “principal place of business is.”

- Issue-spotting: Here, a narrow issue is offered—where is BusinessMart’s principal place of business?
- Rule-recall: A lawyer would recognize that the most relevant rule here comes from the case *Hertz Corp. v. Friend*,⁶ in which the Supreme Court determined “that the phrase ‘principal place of business’ refers to the place where the corporation’s high level officers direct, control, and coordinate the corporation’s activities.”
- Rule-application: Applying this rule to the facts above yields two observations. First, a corporation’s CEO and board members are examples of high level officers referred to in *Hertz* that control and conduct a company. Second, the place where BusinessMart’s high level officers control the company is California, as that is where the CEO and board sit and work.
- Rule-conclusion: Based on the chain of inference spelled out in the application stage, a lawyer would thus conclude that California is BusinessMart’s principal place of business.

The extent to which the outcome of the application and conclusion steps follow each other is dictated by the level of ambiguity in the fact patterns. When the law on a particular question is clear and there is little ambiguity in the facts (as the case in the above example), then the application and conclusion steps point towards the same outcome. Sometimes however, the facts may be unclear or contested, and reasonable minds may differ as the conclusion step. For now, LEGALBENCH focuses entirely on the former setting (unambiguous answers), and all tasks are considered to have objectively “correct” answers.

Other types of reasoning Though IRAC is the most formal framework for legal reasoning, lawyers recognize a variety of skills which are useful to practice of law [47, 77]. For instance, lawyers are often required to exercise interpretive skills, in order to identify the rights, obligations, or limitations of certain legal language (e.g., what a contractual clause may or may not enable). They must also exhibit rhetorical skills, and understand the types of arguments that are made. Though these tasks require the knowledge base and skill set of lawyers, they, arguably, do not always fit neatly within the IRAC framework. Hence, we consider these to be distinct from the examples offered in the previous section.

3.2 Evaluating legal reasoning in large language models

LEGALBENCH identifies six categories of legal reasoning. For each category, we describe how a LLM task may evaluate the typified legal reasoning, using examples from LEGALBENCH.

Issue-spotting LEGALBENCH evaluates issue-spotting through tasks in which an LLM must determine if a set of facts raise a particular set of legal questions, implicate an area of the law, or are relevant to a specific party. Issue tasks evaluate a LLM’s ability to reason over the legal implications of different activities, events, and occurrences.

⁵Diversity jurisdiction gives federal courts the ability to hear cases between parties that are “citizens” of different states.

⁶*Hertz Corp. v. Friend*, 559 U.S. 77 (2010).

An example of an issue-spotting task is the `learned_hands_benefits` task, which requires an LLM to determine (Yes/No) whether a post on a public legal aid forum raises issues related to welfare law (i.e., public benefits or social services). The box below shows how a LLM might be prompted for this task.

Issue-spotting example: `learned_hands_benefits`

Does the post discuss public benefits and social services that people can get from the government, like for food, disability, old age, housing, medical help, unemployment, child care, or other social needs?

Post: “I am currently receiving support from social services, idk why, this is just how my life turned out. They have asked for all of my bank information for the past 12 months. I don’t know what this means. Why would they want that?”

Answer: Yes

Rule-recall LEGALBENCH evaluates rule-recall through tasks which require the LLM to generate the correct legal rule on an issue in a jurisdiction (e.g., the rule for hearsay in US federal court). A rule task can be an open-ended generation task—in which the LLM must generate the text of the rule for a jurisdiction—or a classification task—in which the LLM must determine whether the rule exists in that jurisdiction. Anchoring to jurisdiction is important, as legal rules differ across different jurisdictions. Rule tasks are particularly useful for measuring *hallucinations* [81]. An example of a rule-recall task is `rule_qa`, a question-answer task where questions include asking the model to state the formulations for different legal rules, identify where laws are codified, and general questions about doctrine.

Rule-recall example: `rule_qa`

Question: What are the four requirements for class certification under the Federal Rules of Civil Procedure?”

Answer: Numerosity, commonality, typicality, adequacy

Rule-conclusion LEGALBENCH evaluates rule-conclusion through tasks which require an LLM to determine the legal outcome of a set of facts under a specified rule. LLMs are evaluated purely on whether their predicted outcome is correct. For example, the `ucc_v_common_law` task asks a LLM to determine whether a contract is governed by the Uniform Commercial Code (UCC) or the common law of contracts. The LLM is always provided with the relevant rule, via the prompt (see below).

Conclusion example: `ucc_v_common_law`

The UCC (through Article 2) governs the sale of goods, which are defined as moveable tangible things (cars, apples, books, etc.), whereas the common law governs contracts for real estate and services. For the following contracts, determine if they are governed by the UCC or by common law.

Contract: Alice and Bob enter into a contract for Alice to sell her bike to Bob for \$50. Is this contract governed by the UCC or the common law?

Governed by: UCC

Rule-application LEGALBENCH evaluates rule-application through the same tasks used to measure rule-conclusion. When evaluating rule-application however, we prompt the LLM to provide an explanation of how the rule applies to a set of facts, and evaluate the quality of the generated explanation along two dimensions: (1) whether the explanation is *correct*, and (2) whether it contains *analysis*. Each metric captures a different dimension upon which a particular rule-application may be good.

Correctness corresponds to the criteria that explanations should not contain errors. We focus on five types of errors: misstatements of the legal rule, misstatements of the fact pattern, incorrectly asserting the legal outcome, logic errors, and arithmetic errors. Analysis corresponds to the criteria that explanations should contain inferences from the facts that are relevant under the rule, and illustrate how a conclusion is reached. Consider, for example, an explanation which restates the rule, the fact pattern, and the predicted legal outcome. If the predicted legal outcome is correct, then the explanation in its entirety would be correct, because it contains no error. However, as prior works have noted [71, 30], examples like this are conclusory, and often unsatisfactory in the context of legal work.

To standardize evaluation and enable future work, we have released an “answer guide” for each task used for rule-application, which contains the inferences required for each sample, and describes common modes of errors. All evaluations in LEGALBENCH for rule-application have been performed with respect to this answer-guide.

Table 1 presents an examples of how three generations (corresponding to the Alice/Bob example above) would be evaluated under the above metrics. The first generation is incorrect, because it misstates the rule. The second generation is correct because it contains no falsehoods, but performs no analysis because it does not articulate inferences. The third generation is both correct and contains analysis, because it has no errors, and explicitly mentions an essential inference (e.g., that a bike is a “good”).

Interpretation LEGALBENCH evaluates interpretation through tasks which require the LLM to parse and understand a legal text. Interpretive tasks provide the LLM with a text, and ask the LLM to either extract a relevant piece of information, answer a

Incorrect	Correct, but no analysis	Correct and contains analysis
The contract is for Alice to sell her bike to Bob. The contract is governed by the common law, because all goods are governed by the common law.	The contract is for Alice to sell her bike to Bob. The contract is governed by the UCC, because the UCC governs all goods.	The contract is for Alice to sell her bike to Bob. The contract is governed by the UCC, because a bike is a good and all goods are governed by the UCC.

Table 1: An example of how different generations are evaluated for correctness and analysis.

question, or categorize the text by some property. Interpretive tasks are among the most studied and practically relevant tasks in LEGALBENCH, and many have been taken from actual use-cases. An example of an interpretive task is `cquad_audit_right`, which asks the LLM to determine if a contractual clause contains an “audit right.” An example is shown below:

Interpretation example: `cquad_audit_right`

Does the clause give a party the right to audit the books, records, or physical locations of the counterparty to ensure compliance with the contract?

Clause: “We shall have the right at all times to access the information system and to retrieve, analyze, download and use all software, data and files stored or used on the information system.”

Answer: Yes

Rhetorical-understanding LEGALBENCH evaluates rhetorical-understanding through tasks which require an LLM to reason about legal argumentation and analysis. In these tasks, an LLM is provided with a legal argument (usually excerpted from a judicial opinion), and asked to determine whether it performs a certain function or has a certain property. An example is the `definition_classification` task, in which an LLM must determine if a sentence from a judicial opinion provides a definition of a term.

Rhetorical-understanding example: `definition_classification`

Does the sentence define a term?

Sentence: “To animadvert carried the broader implication of “turn[ing] the attention officially or judicially, tak[ing] legal cognizance of anything deserving of chastisement or censure; hence, to proceed by way of punishment or censure.” 1 Oxford English Dictionary 474 (2d ed.1989).”

Answer: Yes

We emphasize one aspect of LEGALBENCH: IRAC in this work is used as an organizing principle for grouping tasks. On a law exam, a student would be expected to generate an answer which structurally resembles IRAC, where each step builds on the inferences of the previous step [71, 30]. LEGALBENCH tasks, in contrast, each evaluate a single type of legal reasoning. Hence, a task like `learned_hands_benefits` can only be used to evaluate issue-spotting, and not rule-recall. In future work we hope to add tasks which evaluate multiple steps jointly.

4 LEGALBENCH tasks

Appendix F discusses each task in detail, providing a description of the reasoning that each task evaluates, how task data was constructed, task examples, and evaluation protocols. This section provides an overview of LEGALBENCH.

4.1 Construction process

Task sources LEGALBENCH tasks are drawn from three sources. The first source of tasks are existing available datasets and corpora. Most of these were originally released for non-LLM evaluation settings. In creating tasks for LEGALBENCH from these sources, we often significantly reformatted data and restructured the prediction objective. For instance, the original CUAD dataset [62] contains annotations on long-documents and is intended for evaluating extraction with span-prediction models. We restructure this corpora to generate a binary classification task for each type of contractual clause. While the original corpus emphasized the long-document aspects of contracts, our restructured tasks emphasize whether LLMs can identify the distinguishing features of different types of clauses. The second source of tasks are datasets that were previously constructed by legal professionals but never released. This primarily includes datasets hand-coded by legal scholars as part of prior empirical legal projects (e.g., [27]). The last category of tasks are those that were developed specifically for LEGALBENCH, by the authors of this paper. Overall, tasks are drawn from 36 distinct corpora.

Collaborative component In August 2022, we published a call for tasks, describing the goals of the project and its structure [59]. We publicized the project through mailing lists and legal computational conferences. Submitted tasks were vetted for legal correctness and task validity. Task contributors are drawn from diverse professional backgrounds within the law (e.g., academics, practitioners, computational legal researchers) and constitute the authors of this paper.

Infrastructure LEGALBENCH comes with support designed to enable non-law AI researchers to use and study LEGALBENCH tasks. First, each LEGALBENCH task is accompanied by extensive documentation describing how the task is performed, its legal significance, and the construction procedure. The objective of this documentation is to provide AI researchers with a working understanding of the mechanical processes behind each task, for the purposes of better understanding LLM performance. Second, each task is accompanied by a “base” prompt, which contains task instructions and demonstrations. The base prompt is provided to promote replicability and standardization. We anticipate that future research efforts building off of LEGALBENCH will identify higher performing prompts/prompt formats. We intended to update the LEGALBENCH GitHub repository with these prompts as they are discovered.

Limitations We note several limitations of the current LEGALBENCH tasks (additional limitations are noted in Appendix B). First, when this project began, most LLM context-windows were constrained to a few pages of text. As a result, the initial round of LEGALBENCH tasks does not involve longer documents. We hope to include such tasks in future work, particularly as recent technical developments have resulted in significantly longer context windows [42, 54, 43, 109]. Second, LEGALBENCH’s tasks focus on legal reasoning questions with objectively correct answers. LEGALBENCH is thus not helpful for evaluating legal reasoning involving degrees of correctness or tasks where “reasonable minds may differ.” Third, LEGALBENCH only considers English language tasks, is skewed towards certain jurisdictions (American law), and certain areas of the law (contracts). Thus, the current iteration of the benchmark limits inferences regarding how LLMs may generalize to legal tasks involving other jurisdictions. As we continue to solicit and incorporate contributions to LEGALBENCH, we hope to add tasks addressing these limitations. Finally, LEGALBENCH evaluates IRAC abilities independently, while law exams and other legal work requires lawyers to generate outputs which follow IRAC in a multi-hop matter (i.e., each aspect is applied to the same fact pattern).

4.2 Dimensions of variation

Task structure All LEGALBENCH tasks contain at least 50 samples, with an average task size of 563 samples (Appendix D.4). These tasks are comparable in size to those used in benchmarking efforts like BigBench [130], HELM [80] or RAFT [1]. LEGALBENCH tasks also span different formats: multiple-choice questions (35 tasks), open-generation (7 tasks), binary classification (112 tasks), and multi-class/multi-label classification (8 tasks).

Reasoning types and legal domains LEGALBENCH provides tasks for each of the reasoning categories discussed above: rule-recall (5 tasks), issue-spotting (16 tasks), rule-application (16 tasks), rule-conclusion (16 tasks), interpretation (119 tasks), and rhetorical-understanding (10 tasks). Tasks are predominantly drawn from areas of law implicating civil matters, including contracts (58 tasks), civil procedure (8 tasks), evidence law (1 task), and corporate law (58 tasks). The skew towards interpretation tasks and tasks from contract law can be explained by the ubiquity of legal documents from these areas (e.g., contracts, terms-of-service agreements, disclosures, and etc.) and their immediate commercial implications [62, 76].

Language variation Legal language is highly heterogeneous, varying in sentence structure, vocabulary, and rhetorical style across different legal areas and document types [60]. This poses a distinct challenge for LLMs, which are extremely sensitive to structure of input text and the vocabulary used [80]. LEGALBENCH tasks are drawn from a diverse set of legal language types, thus enabling researchers to study performance variation across different categories of legal text. Specifically, LEGALBENCH encompasses tasks with language drawn from plain English (32 tasks), legal opinions (11 tasks), merger agreements (34 tasks), contracts (55 tasks), statutory text (3), and other sources.

4.3 Tasks

We offer a brief summary of the tasks present in each reasoning category.

Issue-spotting There are 17 issue-spotting tasks. 16 tasks are derived from the “Learned Hands” Dataset (Section F.14). Each of these tasks is a binary classification task, in which the LLM must determine if a post from */r/legaladvice* implicates a particular domain of law (e.g., immigration). The last task is the *corporate_lobbying* task (Section F.7), which requires determining if a legislative bill has legal implications for a described company.

Rule-recall There are 5 rule-recall tasks. Two tasks require an LLM to either generate the citation for a particular legal quote, or identify if a candidate citation is correct (Section F.3). The remaining three tasks are:

- *rule_qa*, in which the LLM must generate the text of different legal tests and identify where they’re codified (Section F.25).
- *international_citizenship_questions*, in which the LLM must answer yes/no questions about citizenship requirements in different countries (Section F.13).

- `nys_judicial_ethics`, in which the LLM must answer yes/no questions corresponding to different ethical rules under the guidance provided by the New York State Advisory Committee on Judicial ethics (Section F.17).

Rule-application and rule-conclusion There are 12 tasks used for both rule-application and rule-conclusion.

- Six tasks evaluate an LLM’s ability to apply the diversity jurisdiction test to information about plaintiff and defendant citizenships and the amount-in-controversy for different claims (Section F.9). This requires both arithmetic and logical reasoning. The simplest (`diversity_1`) involves one plaintiff, one defendant, and one legal claim. The most complex (`diversity_6`) involves two plaintiffs, two defendants, and two claims against each defendant.
- `abecrombie` evaluates an LLM’s ability to apply the *Abercrombie* test to classify how distinctive a product/service name is for a particular product/service (Section F.1).
- `hearsay` evaluates an LLM’s ability to identify—given a particular piece of evidence and an issue being litigated—whether the evidence would count as hearsay for that issue (i.e., an out-of-court statement introduced to prove the truth of the matter asserted) (Section F.11).
- `personal_jurisdiction` evaluates an LLM’s ability to identify when a court in a particular forum may exercise personal jurisdiction over a defendant, given basic facts about the defendant’s place of domicile, their interactions with the state, and the claims brought against them by plaintiffs (Section F.21).
- `successor_liability` evaluates an LLM’s ability to identify the potential successor liability exceptions present in fact patterns describing a sale of assets from one company to another (Section F.29).
- `telemarketing_sales_rule` evaluates an LLM’s ability to identify whether the representations made by a company covered under the Telemarketing Sales Rule violate either 16 C.F.R. § 310.3(a)(1) and 16 C.F.R. § 310.3(a)(2), which outline a series of specific telemarketing sales practices defined as “deceptive” (Section F.31).
- `ucc_v_common_law` evaluates an LLM’s ability to determine whether a particular contract is covered by the Uniform Commercial Code (UCC) or the common law, given information about the contract (Section F.33).

Interpretation There are 118 interpretation tasks.

- `consumer_contracts_qa`, which evaluates an LLM’s ability to determine the rights/obligations imposed by terms of service clauses from popular websites (Section F.5).
- `contract_qa`, which evaluates an LLM’s ability to identify different types of contractual provisions.
- 14 tasks designed from the ContractNLI dataset [74]. Each task evaluates an LLM’s ability to identify whether a candidate contract excerpt adheres to a task-specific assertion (Section F.6).
- 38 binary-classification tasks designed from the CUAD dataset [62]. Each task evaluates an LLM’s ability to identify whether a candidate contractual clause is of a certain type (Section F.4.1).
- `insurance_policy_interpretation`, which evaluates an LLM’s ability to determine whether a particular claim is covered by an insurance policy (Section F.12).
- `jcrew_blocker`, which evaluates an LLM’s ability to identify whether a particular loan clause is a J.Crew Blocker provision (Section F.4.2).
- 34 tasks from the MAUD dataset [142], which evaluates an LLM’s ability to answer multiple-choice questions about the content of excerpts from merger-agreements (Section F.16). Each task corresponds to a different question.
- 9 tasks from the OPP-115 dataset [149], each of which evaluates an LLM’s ability to determine whether a privacy policy clause discusses a particular issue (Section F.18). Each task is a binary classification task corresponding to a different issue.
- `privacy_policy_entailment` [161], which evaluates an LLM’s ability to answer entailment questions from privacy policies (Section F.22).
- `privacy_policy_qa` [112], which evaluates an LLM’s ability to determine if a clause from a privacy policy contains the answer to a particular question (Section F.23).
- 2 tasks designed from the SARA dataset [65], which evaluate an LLM’s ability to interpret and apply sections of the tax-code (Section F.26).
- 10 tasks which evaluate an LLM’s ability to identify when a supply chain disclosure discusses or describes a particular type of information (Section F.30). Each task corresponds to a different disclosure objective.
- `unfair_tos` [82], which evaluates an LLM’s ability to classify clauses from terms of service agreements into one of multiple categories (Section F.4.3).

Rhetorical-understanding There are 10 tasks which evaluate rhetorical-understanding.

- `canada_tax_court_outcomes` evaluates an LLM’s ability to identify the outcome of a tax court decision, based on the text of the decision (Section 16).
- 2 tasks evaluate an LLM’s ability to (1) identify sentences from US Supreme Court opinions which define a term, and (2) extract that term (Section F.8).
- `function_of_decision_section` evaluates an LLM’s ability to identify the function that an excerpt of a legal opinion has (e.g., statement of rule) (Section F.10).
- `legal_reasoning_causality` evaluates an LLM’s ability to identify when an excerpt of a court’s opinion relies on statistical evidence (Section F.15).
- `oral_argument_question_purpose` evaluates an LLM’s ability to identify the purpose that a particular question (from Supreme Court oral arguments) plays (Section F.19).
- `overruling` [157] evaluates an LLM’s ability to identify when a sentence from a judicial opinion overrules a previous case (Section F.20).
- `scalr` evaluates an LLM’s ability to assess which holding statement (amongst several options) best answers a provided legal question.
- 2 tasks evaluate an LLM’s ability to identify whether excerpts of judicial reasoning rely on certain textualist tools (Section F.32). Each task corresponds to a different tool.

5 Results

We use LEGALBENCH to conduct a three-part study.

- In the first part (Section 5.2), we conduct a sweeping evaluation of 20 LLMs from 11 different families, at four different size points. We use this study to make initial observations on performance differences across families, the role of model size, and the gap between open-source and commercial LLMs.
- In the second part (Section 5.3), we show how LEGALBENCH can be used to conduct in-depth evaluations of models. To illustrate, we use LEGALBENCH to highlight similarities and differences in the performance of three popular commercial models: GPT-4, GPT-3.5, and Claude-1.
- In the final part (Section 5.4), we show how LEGALBENCH can support the development of law-specific LLM methods. We focus on prompting, and conduct a series of experiments that begin to surface tradeoffs and challenges with regards to guiding LLMs towards certain tasks.

Ultimately, our study here serves to illustrate the types of analyses that LEGALBENCH enables, and highlight potential directions for future work.

5.1 Setup

5.1.1 Models

Commercial models We study three commercial API-access models. From the OpenAI GPT family, we study GPT-3.5 [14] (text-davinci-003) and GPT-4 [98]. Results from these models were retrieved between May and August of 2023. From the Anthropic family, we study Claude-1 (v1.3) [3]. Results from this model were retrieved in July of 2023. These models are believed to be large (hundreds of billions of parameters), though exact details on their architecture and training process are unknown. It is thus possible that some LEGALBENCH tasks leaked into pretraining data. Details on the extent to which different LEGALBENCH tasks have been previously made available online can be found in Appendix D.

Open-source models We study 17 open-source models at three different size points: 3B parameters, 7B parameters, and 13B parameters. All inference was performed on two-GPU GCP 40GB A100s, using the Manifest library [99]. HuggingFace links for each model are provided in Appendix G.

- From Together, we study three models: Incite-Instruct-7B, Incite-Base-7B, and Incite-Instruct-3B [35, 135].
- From Meta’s OPT family, we study three models: OPT-2.7B, OPT-6.7B, and OPT-13B [156].
- From TII’s Falcon family, we study Falcon-7B-Instruct [2, 105].
- From MosaicML’s MPT family, we study MPT-7B-8k-Instruct [131].
- From LMSYS’ Vicuna family, we study Vicuna-7B-16k and Vicuna-13B-16k [26].
- From Google’s FLAN-T5 family, we study Flan-T5-XL (3B parameters) and Flan-T5-XXL (11B parameters) [32].
- From Meta’s LLama-2 family, we study LLaMA-2-7B, and LLaMA-2-13B [136].

- From the Wizard family, we study WizardLM-13B [152].
- From the BigScience BLOOM family, we study BLOOM-3b and BLOOM-7B [118].

Future work Our selected LLMs represent only a sample of the models available. For instance, we do not evaluate LLMs larger than 13B parameters, which have been observed to perform well [9]. Studied LLMs are also “general domain,” in that we don’t find evidence that any were specifically customized to perform well on legal text.⁷ In future work we hope to expand our evaluation to a broader set of LLMs.

5.1.2 Prompts

We designed a prompt for each task by manually writing instructions for the task, and selecting between zero and eight samples from the available train split to use as in-context demonstration. The number of samples selected depended on the availability of data and the sequence length of samples (Appendix G.2). For instance, the inputs to the Supply Chain Disclosure tasks are disclosure statements between 1-2 pages long, making the inclusion of multiple demonstrations infeasible. For application evaluation, we augmented the prompt with an instruction for the LLM to explain its reasoning.

We used the same prompts across all LLMs with one exception. In contrast to the OpenAI and open-source LLMs, Anthropic recommends specific prompting formats when using Claude.⁸ This includes surrounding in-context samples with `<example>/example>` tags, and adding instructions specifying the output space. We observed that failing to adhere to these guidelines led Claude to generate text which made extracting a prediction challenging. Therefore, when prompting Claude, we added example-tags to the in-context demonstrations and instructions specifying the prediction space (e.g., “Reply with either: generic, descriptive, suggestive, arbitrary, fanciful”).

LLM outputs were generated using next-token generation at a temperature of 0.0. For classification/extraction tasks, we terminated at a new-line token. For `rule_qa` and all application tasks except `diversity_jurisdiction_6` we generated 150 tokens. For `diversity_jurisdiction_6` we generated 300 tokens.

We believe there is significant scope for improving and refining prompts on LEGALBENCH. Hence, our results here provide a lower-bound on performance, as better prompts may elicit higher scores. Our prompts correspond to what we believe would be reasonable, based on experience with prompt engineering in other settings, and the guidance provided by model developers. We make all prompts available as a starting point for future work on LEGALBENCH.

5.1.3 Evaluation

Classification tasks are evaluated using “exact-match” (following HELM [80]). Because some tasks contain significant label imbalances, we use balanced-accuracy as a metric. For extraction tasks, we perform normalization on generated outputs to account for differences in tense/casing/punctuation. A few tasks (e.g., `successor_liability` and `ssla_individual_defendants`) requires the LLM to produce multiple classes or extracted terms per instance. For these, we evaluate using F1. Appendix E provides more details.

Rule-application tasks were evaluated manually by a law-trained individual, who analyzed LLM responses for both correctness and analysis.⁹ This type of manual evaluation is consistent with previous works evaluating LLM generations in the legal domain [30, 71]. As rule-application requires LLMs to generate “explanations” detailing legal reasoning—a capability primarily exhibited by larger models—we only evaluated GPT-4, GPT-3.5, and Claude-1. `rule_qa` was also manually evaluated by a law-trained individual. Appendix E provides more details on our approach to manual grading. All manual evaluation was performed with reference to a grading guide, which we additionally make available.

5.2 Performance trends

Table 2 provides the average task performance for all 20 models in five reasoning categories (issue-spotting, rule-recall, rule-conclusion, interpretation, and rhetorical-understanding). The first block of rows corresponds to large commercial models, the second block corresponds to models in the 11B-13B range, the third block corresponds to models in the 6B-7B range, and the final block corresponds to models in the 2B-3B range. Table 3 provides the average task performance for the three large models on rule-application. Appendix G provides full results for each model on each task.

Overall, we find significant variation in performance across tasks, suggesting that LEGALBENCH captures a diverse spectrum of difficulty (Appendix G). These results emphasize that assessments of LLM capabilities for legal applications must be made on a task-by-task basis, and informed by the nuances of specific tasks. While certain types of tasks appear beyond the scope of current-day LLMs, others seem more within reach. In this section, we offer preliminary observations on performance trends across model size, family, and reasoning categories.

⁷We note that as of July 2023, we were unable to identify public law-specific English large language models to evaluate.

⁸<https://docs.anthropic.com/claude/docs/introduction-to-prompt-design>

⁹For the six diversity jurisdiction tasks, we sampled 30 instances from each task. For all other rule-application tasks, we manually evaluated the entirety of the dataset.

LLM	Issue	Rule	Conclusion	Interpretation	Rhetorical
GPT-4	<u>82.9</u>	<u>59.2</u>	<u>89.9</u>	<u>75.2</u>	<u>79.4</u>
GPT-3.5	60.9	46.3	78.0	72.6	66.7
Claude-1	58.1	57.7	79.5	67.4	68.9
Flan-T5-XXL	<u>66.0</u>	36.0	<u>63.3</u>	64.4	<u>70.7</u>
LLaMA-2-13B	50.2	37.7	<u>59.3</u>	50.9	54.9
OPT-13B	52.9	28.4	45.0	45.1	43.2
Vicuna-13B-16k	34.3	29.4	34.9	40.0	30.1
WizardLM-13B	24.1	<u>38.0</u>	62.6	50.9	59.8
BLOOM-7B	50.6	24.1	47.2	42.8	40.7
Falcon-7B-Instruct	51.3	25.0	52.9	46.3	44.2
Incite-7B-Base	50.1	<u>36.2</u>	47.0	46.6	40.9
Incite-7B-Instruct	<u>54.9</u>	35.6	52.9	<u>54.5</u>	<u>45.1</u>
LLaMA-2-7B	50.2	33.7	<u>55.9</u>	47.7	47.7
MPT-7B-8k-Instruct	54.3	25.9	<u>48.9</u>	42.1	44.3
OPT-6.7B	52.4	23.1	46.3	48.9	42.2
Vicuna-7B-16k	3.9	14.0	35.6	28.1	14.0
BLOOM-3B	47.4	20.6	45.0	45.0	36.4
Flan-T5-XL	<u>56.8</u>	<u>31.7</u>	<u>52.1</u>	<u>51.4</u>	<u>67.4</u>
Incite-3B-Instruct	51.1	26.9	47.4	49.6	40.2
OPT-2.7B	53.7	22.2	46.0	44.4	39.8

Table 2: Average performance for each LLM over the different LEGALBENCH categories. The first block of rows corresponds to large commercial models, the second block corresponds to models in the 11B-13B range, the third block corresponds to models in the 6B-7B range, and the final block corresponds to models in the 2B-3B range. The columns correspond to (in order): issue-spotting, rule-recall, rule-conclusion, interpretation, and rhetorical-understanding. For each class of models (large, 13B, 7B, and 3B), the best performing model in each category of reasoning is underlined.

LLM	Correctness	Analysis
GPT-4	<u>82.2</u>	<u>79.7</u>
GPT-3.5	58.5	44.2
Claude-v1	61.4	59.0

Table 3: Average performance for the large LLMs on rule-application tasks.

Parameter count Within LLM families, we observe that larger models usually outperform smaller models. For instance, Flan-T5-XXL (11B parameters) outperforms Flan-T5-XL (3B parameters) on average across all five reasoning categories, and LLaMA-2-13B outperforms LLaMA-2-7B on average across four reasoning categories. Notably, the margin of the gap varies across LLM families and reasoning categories. For instance, on rule-recall, the 7B Incite-Instruct model outperforms the 3B Incite-Instruct model by almost 10pts, while the 6.7B OPT model outperforms the 2.7B OPT model by less than 1pt. We additionally note that the largest LLM (GPT-4) outperforms virtually all other models.

Variation across families Even for LLMs of the same size, we find considerable differences in performance. For instance, we observe significant gaps in performance between Flan-T5-XXL (11B parameters) and Vicuna-13B-16k (13B parameters), across all reasoning categories. This suggests, unsurprisingly, that the choice of pretraining data, regime of instruction-tuning, and architecture play an important role in determining performance, and that certain configurations may be better aligned for LEGALBENCH tasks. Interestingly, we observe that such choices may affect which types of reasoning categories LLMs appear to perform well at. For instance, we observe that WizardLM-13B performs worse than all peers on issue-spotting tasks, best on rule-recall tasks, and nearly matches the performance of the best-performing peer on rule-conclusion tasks. Comparing Incite-7B-Instruct to Incite-7B-Base also provides insight into the effect of instruction-tuning across different categories, at one size point (7B parameters). We observe that instruction-tuning improves performance on four categories (issue-spotting, rule-conclusion, interpretation, and rhetorical-understanding), and worsens performance on rule-recall.

We additionally find that family-specific trends appear to hold across different size points. For instance, the Flan-T5 models outperform all others at both the 3B and 13B scale, while the Vicuna models appear to underperform competitors at both the 7B and 13B scale. We attribute the Vicuna models’ low performance to their frequency tendency to generate poorly-formed outputs, which

did not map to the expected verbalizer tokens (e.g., blank spaces, random characters, etc.).¹⁰ This could possibly be attributed to the type of data used to fine the model (e.g., user-conversation), although more in-depth experimentation is necessary.

The gap between open-source and commercial models Finally, we find evidence that open-source models are capable of performance that matches or exceeds certain commercial models. For instance, Flan-T5-XXL outperforms GPT-3.5 and Claude-1 on two categories (issue-spotting and rhetorical-understanding), despite the relative gap in parameter count. Notably, the gap between closed and open-source models is largest for the rule-conclusion category. Amongst LEGALBENCH tasks, rule-conclusion tasks most like the other types of multi-step/common-sense reasoning tasks where commercial LLMs have been found to perform well.

5.3 Comparing GPT models

This section provides a more in-depth study of performance, focusing on the three commercial models (GPT-4, GPT-3.5, and Claude-1). The purpose of this section is to illustrate how LEGALBENCH enables fine-grained analysis of LLM performance. In particular we highlight how LEGALBENCH can provide more rigorous empirical support for anecdotal observations arising out of the legal community’s use of these models, and explain performance differences between models.

5.3.1 Issue-spotting

We first consider average model performance across all issue-spotting tasks. We observe that GPT-4 outperforms GPT-3.5 and Claude-1 (both at $p < 0.001$).¹¹ In absolute terms, issue tasks present the largest gap in performance between GPT-4 and other closed-API models, with an absolute margin of 20+ points. GPT-3.5 and Claude-1, in contrast, appear to match each other in performance, separated by an average gap of only 2 points. We additionally find that the open-source models perform poorly here. On 9 tasks, Incite-Base collapses to predicting a single class for all samples.

We note one limitation to our results: because 16/17 of our issue-spotting tasks are drawn from one source (Learned Hands data), average issue performance is skewed by properties of the Learned Hands data distribution (i.e., user-generated questions). For instance, though GPT-3.5 outperforms Claude-1 on 12/16 Learned Hands tasks, Claude-1 outperforms GPT-3.5 on the one non-Learned Hands task (`corporate_lobbying`). Despite the skew, we observe that these tasks appear to vary in difficulty. While GPT-4’s balanced-accuracy on `learned_hands_torts` is only 70.6%, on three tasks—`learned_hands_immigration`, `learned_hands_traffic`, and `learned_hands_estate`—it scores $> 95\%$.

5.3.2 Rule-recall

We first consider average model performance across all rule-recall tasks. While GPT-4 outperforms GPT-3.5 ($p < 0.05$), we surprisingly find that Claude-1 also outperforms GPT-3.5 ($p < 0.05$), and appears almost on par with GPT-4. Moreover, Claude-1 outperforms GPT-4 on three tasks: `rule_qa`, `international_citizenship_questions`, and `nys_judicial_ethics`. This is the only task category where Claude-1 provides performance comparable to GPT-4. Because little is known regarding the architecture and training processes for these models however, it is difficult to explain why this is the case.

Because rules/laws can be analogized to law-specific “facts,” rule-recall tasks are similar to general domain LLM tasks designed to measure “hallucination.” There, an extensive literature has documented the propensity for LLMs to both generate factually incorrect information, and answer fact-based questions incorrectly [80, 106]. Our results align with the primary findings of that literature. For example, we observe that the small open source models perform considerably worse than the larger models, consistent with the observation that model size plays an important role in fact-retention. Overall, performance on the rule-recall tasks also lend additional empirical support to more anecdotal reports—from the legal community—regarding how LLMs often misstate the law or cases [114].

5.3.3 Rule-application

Application tasks evaluate whether LLMs can explain how a legal rule applies to a set of facts, and verbalize the necessary inferences. With respect to correctness, we observe that GPT-4 outperforms both GPT-3.5 ($p < 0.01$) and Claude-1 ($p < 0.01$). Across LLMs, we find that variation in performance across tasks is consistent with subjective impressions of task difficulty. For instance, performance on `diversity_jurisdiction_1` (an easy task requiring a model to determine if an amount is greater than \$75k and if the plaintiff and defendant are from different states) is much higher than performance on `successor_liability` (a harder task requiring a model to identify multiple successor liability exceptions in a fact pattern describing a complex transaction).

We observe that LLM generations may be incorrect in many different ways. On the Diversity Tasks, LLMs sometimes perform incorrect arithmetic operations or mathematical comparisons (i.e., stating that \$50,000 is greater than \$75,000). On `telemarketing_sales_rule` in contrast, LLMs will cite to an incorrect portion of the rule. For instance, a generation may explain that certain conduct by a telemarketer runs afoul of the rule because the telemarketer failed to make a mandatory disclosure (16 CFR § 310.3(a)(1)), but cite to the portion of the rule prohibiting misrepresentations (16 CFR § 310.3(a)(2)). Examples of other types of incorrect generations can be found in Table 4.

¹⁰In further experimentation, we found that writing prompts using the “### Human:” and “### Assistant:” templates did not appear to help.

¹¹Statistical significance is computed using a paired t -test over the tasks in the category.

Fact Pattern (Task)	LLM generation	Why is it incorrect?
Sophia is from Nebraska. Ava is from Washington. Sophia sues Ava for legal malpractice for \$34,000. (diversity_1)	“No. The parties are not diverse because Sophia and Ava are both from different states. The AiC is too small because it is less than 75k. Because the AiC requirement is not met there is no jurisdiction.” (GPT-3.5)	While GPT-3.5 correctly identifies that there is no jurisdiction because the AiC is less than \$75k (and thus too small), it fails to recognize that the parties (Sophia and Ava) are diverse because they are from different states.
The mark "Trim" for nail clippers. (abercrombie)	“The mark trim for nail clippers would be considered a suggestive mark. It suggests a characteristic or quality of the product which is that the nail clippers are used to trim ones nails. It implies this purpose and function without directly describing it so trim for nail clippers would be classified as a suggestive trademark.” (Claude-1)	Claude-1’s assertion that the mark “implies this purpose and function without directly describing it” misses the fact that “Trim” directly describes, rather than implies, what nail clippers are used for (i.e., the <i>trim</i> nails). Thus, the mark is actually descriptive, and not suggestive.
Ana is a lawyer who resides in Texas. While visiting Louisiana, she meets David, who runs a bike shop. She buys a bike from him and uses it to bike back to Texas. Right after she crosses the border, the bike seat explodes, injuring Ana. Ana sues David in Louisiana. (personal_jurisdiction)	“Yes. There is personal jurisdiction over David in Louisiana. Explanation. 1. David is domiciled in Louisiana so there is general jurisdiction over him in Louisiana. 2. David has sufficient minimum contacts with Louisiana such that he has availed himself of the privileges of conducting business in Louisiana. He operates a bike shop in Louisiana so he purposefully avails himself of doing business in the state.. 3. Ana’s claim arises out of Davids contacts with Louisiana she bought the defective bike from David’s bike shop in Louisiana so the claim is directly related to David’s business contacts in the state. 4. Requiring David to litigate in Louisiana would not offend traditional notions of fair play.” (Claude-1)	The LLM states that David is domiciled in Louisiana, when the facts make no mention of this.

Table 4: Examples of generations for rule-application tasks which are *incorrect*. The last column provides analysis for why the generation is incorrect.

With respect to analysis, we observe that GPT-4 again outperforms both GPT-3.5 ($p < 0.01$) and Claude-1 ($p < 0.01$). Explanations which failed to exhibit analysis can be grouped into several categories. First, some generations will contain just a prediction as to the legal outcome, without an explanation (even when the LLM has been prompted to generate one). The same prompt—applied to other samples in the dataset—will elicit explanations containing analysis. Second, we observe a tendency for LLMs to sometimes generate explanations which merely restate the facts and legal rule, without actually offering an explanation for how the outcome is reached. Examples of such instances are provided in the table below.

5.3.4 Rule-conclusion

Rule-conclusion evaluates on the same tasks as rule-application, but only requires the LLM to generate a prediction as to the outcome, and not an explanation. We observe that GPT-4 once again outperforms GPT-3.5 ($p < 0.01$) and Claude-1 ($p < 0.01$). Claude-1 and GPT-3.5 appear approximately level on performance.

The rule-conclusion tasks offer a heuristic for characterizing the types of legal inferences LLMs are capable of or struggle with. In particular, several of these tasks organize samples into *slices*, where the samples contained within a slice all represent a similar type of fact pattern, and thus interact with the legal rule in a comparable way. For instance, the *hearsay* task contains a slice corresponding to “non-verbal hearsay.” This slice contains fact patterns where an individual communicates something non-verbally (e.g., pointing), thus qualifying their conduct as a “statement” under the hearsay rule. In order to make accurate predictions on this slice, an LLM must recognize that (1) the hearsay rule applies to non-verbal communicative conduct, and (2) the non-verbal conduct in these fact patterns is communicative.

Though slices are small—and thus not intended for rigorous statistical analysis—they provide some intuition as to the source of GPT-4’s improvement over GPT-3.5, and the overall areas of strength and weakness for both models. On the *hearsay* task for instance (Table 6), the difference between GPT-4 and GPT-3.5 appears primarily attributable to improvements over the slices corresponding to non-verbal hearsay and statements made in court. In looking across slices moreover, it’s clear that some are comfortably within the realm of model capabilities (e.g., non-assertive conduct), while others (e.g., not introduced to prove the truth of the matter asserted) still pose a considerable challenge.

Fact Pattern (task)	LLM generation	What analysis is missing?
Martha makes special pinwheel hats in her home in Salem, Oregon. She only markets her hats on the west coast (California, Oregon, and Washington). Alex, a citizen of Maine, purchases one of the hats while visiting Salem. When Alex returns to Maine, the pinwheel comes off and seriously cuts Alex. Alex sues Martha for damages in Washington. (personal_jurisdiction)	“No. There is not personal jurisdiction.” (Claude-1)	Claude-1 does not provide any explanation for <i>why</i> there is no personal jurisdiction, so the explanation does not contain analysis.
On the issue of whether Henry negligently hit Kelly with his car, Kelly’s statement, “a witness told me after the crash that Henry was on his cell phone while driving. (hearsay)	“Yes, there is hearsay in this situation. The reasoning is as follows:. 1. The statement in question is an out-of-court statement, as it was made by a witness who is not testifying in court.. 2. The statement is being introduced to prove the truth of the matter asserted, which is that Henry was on his cell phone while driving.. 3. Since the statement is an out-of-court statement being introduced to prove the truth of the matter asserted, it is considered hearsay.” (GPT-4)	GPT-4 does not explain how the truthfulness of the out-of-court statement (whether or not Henry was on his phone) relates to the purpose of its introduction (proving/disproving that Henry negligently drove).

Table 5: Examples of generations for rule-application tasks which do not contain analysis. The last column explains why the generation is deficient.

Slice	Slice description	GPT-3.5	GPT-4
Non-assertive conduct ($n = 19$)	The fact pattern describes conduct which is non-communicative and therefore not hearsay.	100%	100%
Statement made in-court ($n = 14$)	The fact pattern describes a statement that was made in court and therefore not hearsay.	57%	93%
Standard hearsay ($n = 29$)	The fact pattern describes traditional hearsay (out-of-court statement introduced to prove the truth of the matter asserted).	97%	97%
Non-verbal hearsay ($n = 12$)	The fact pattern describes non-verbal communicative conduct that qualifies as hearsay.	33%	75%
Not introduced to prove truth ($n = 20$)	The fact pattern describes a statement <i>not</i> introduced to prove the truth of the matter asserted, which is therefore not hearsay.	25%	45%

Table 6: Comparison between GPT-3.5 and GPT-4 on hearsay slices. Accuracy is reported for each slice.

Another example is provided by the *abercrombie* task, in which an LLM must determine the relationship between a product and a potential trademark name, by classifying the product-name pair into one of five categories recognized by courts: generic, descriptive, suggestive, arbitrary, and fanciful. Loosely, these categories measure how *distinctive* a product name is for a product, with generic being the least distinctive, and fanciful being the most distinctive. Just as with *hearsay*, comparing LLM performance on each of these categories provides insight into the relative areas of improvement (Table 7). Here, GPT-4’s improved overall performance appears most attributable to performance on marks which are suggestive or arbitrary. However, GPT-4 still makes a number of errors for both categories. Interestingly, performance on descriptive marks is consistent between both models.

5.3.5 Interpretation

On the interpretation tasks, we find that on average GPT-4 outperforms GPT-3.5 ($p < 0.01$), and GPT-3.5 outperforms Claude-1 ($p < 0.01$). Here, the larger API-models are highly performant on tasks which involve binary classification over short clauses. Averaged across the 38 CUAD tasks (contract clauses), for instance, GPT-4, GPT-3.5, and Claude-1 all have a balanced-accuracy $\geq 88\%$. And on *proa* (statutory clauses), both GPT-4 and GPT-3.5 have a balanced-accuracy $\geq 90\%$. Notably, performance degrades on tasks which contain longer text sequences or involve multi-class classification. On the Supply Chain Disclosure tasks for instance—in which LLMs must classify disclosures which are 1-2 pages in length—the average balanced-accuracy of the large commercial models ranges between 74-75%. And on the MAUD tasks—which require answering multiple choice questions about merger deals—the average balanced-accuracy of GPT-4 drops to 47.8% accuracy.

Mark	Mark description	GPT-3.5	GPT-4
Generic ($n = 19$)	The name connotes the basic nature of the product/service.	94%	100%
Descriptive ($n = 19$)	The name identifies a characteristic or quality of the product/service.	73%	72%
Suggestive ($n = 20$)	The name suggests, rather than describes, a characteristic of the product/service.	38%	70%
Arbitrary ($n = 18$)	The name is a real world but has no relation to the product/service.	41%	82%
Fanciful ($n = 19$)	The name is a made-up word.	84%	100%

 Table 7: Comparison between GPT-3.5 and GPT-4 on `abercrombie` categories. Accuracy is reported for each slice.

5.3.6 Rhetorical-analysis

On average across all rhetorical-understanding tasks, we find that GPT-4 outperforms both GPT-3.5 ($p \leq 0.05$) and Claude-1 ($p \leq 0.05$). We note several results. First, on `definition_extraction`—which requires a LLM to extract the term defined by a sentence taken from a Supreme Court opinion—Incite-Base almost equals GPT-4 in performance (80.6% accuracy to 81.8%). Second, nearly all evaluated models struggle on two tasks requiring LLMs to label the legal “roles” played by either a question or excerpt from an opinion (`function_of_decision_section` and `oral_argument_question_purpose`). Notably, both tasks require the LLM to classify text into one of six or more categories

5.4 Prompt engineering strategies

Finally, we illustrate—through a series of micro-studies—how LEGALBENCH can be used to explore different aspects of prompt-engineering for LLMs in legal settings. We focus on three questions:

1. Can LLMs rely on their latent knowledge of a rule for rule-conclusion tasks?
2. Does simplifying task descriptions to plain language affect performance?
3. Are LLMs sensitive to the choice of in-context demonstrations?

Reliance on latent knowledge When prompting for general-domain tasks like sentiment or topic classification, prompt-engineers will often rely on the LLM’s latent knowledge of the task [5]. In topic classification for instance, a prompt may use the instructions to label whether a news article is about “sports,” without offering a detailed description of what “sports” refers to or encompasses. Such a description is not necessary, because general-domain terms like “sports” appear frequently in LLM training corpora, and LLMs can learn from these occurrences what general-domain terms mean. Prompting for legal tasks, however, may require a different strategy. Because legal terms occur less frequently in general domain training corpora, legal prompting may require practitioners to provide additional background information. For example, a general domain LLM may not know what the requirements for diversity jurisdiction are, because diversity jurisdiction is not as commonly discussed in pretraining corpora.

We explore this question through a study of rule-conclusion tasks. For a selection of these tasks, we evaluate GPT-3.5 with two zero-shot prompts: a reference-based prompt and a description-based prompt. In the reference prompt, the task instructions merely state the rule to be applied, i.e., “Determine if the following fact patterns give rise to diversity jurisdiction.” In the description-based prompt, the instructions provide an explicit description of the rule, i.e., “Diversity jurisdiction exists when there is (1) complete diversity between plaintiffs and defendants, and (2) the amount-in-controversy (AiC) is greater than \$75k.” By comparing performance between the reference and description prompt, we can measure whether providing a description of the rule in the prompt provides additional performance boost over the LLM’s latent knowledge of the rule.

Figure 1 provides a comparison for the different prompts. Interestingly, we find considerable variation across tasks. On tasks like `abercrombie`, `ucc_v_common_law`, `diversity_2`, and `diversity_4`, description prompts appear to offer significant increase in performance. On the other tasks, performance is approximately the same (or even worse). We identify two possible explanations for diverging results across tasks. First, on certain tasks, subsets of fact-patterns are too challenging for LLMs like GPT-3.5, and description-based prompts do not provide sufficient guidance for LLMs to reason through those fact patterns. Second, legal rules may be described to varying extents within pretraining corpora. Hence, tasks where we observe performance improvements from description-based prompting may correspond to rules which occur less frequently in pretraining data.

Plain language descriptions of tasks Next, we examine the extent to which domain specialization in the language of the prompts affects performance. Like experts in other specialized domains, lawyers have developed their own language (i.e., “legalese”), which forms the basis for most legal writing and communication. It is unclear whether—in interacting with large language models through prompting—lawyers should continue to rely on formalistic legal language, or instead use simpler plain language. While most large language models are “general domain” and thus less specialized to legalese, formalistic legal language is more precise, and may thus induce more accurate behavior from the model.

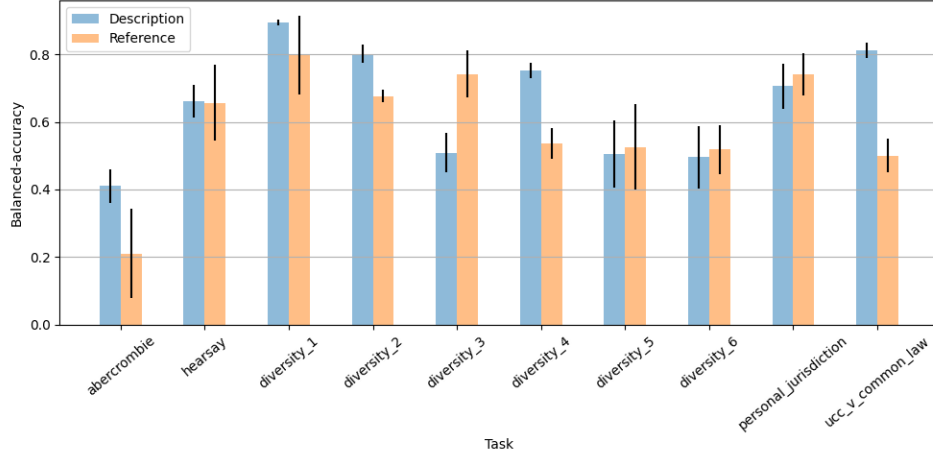


Figure 1: We compare performance of prompts which describe the legal rule to be applied (“description”) against prompts which reference the legal rule to be applied (“reference”). Error bars measure standard error, computed using a bootstrap with 1000 resamples.

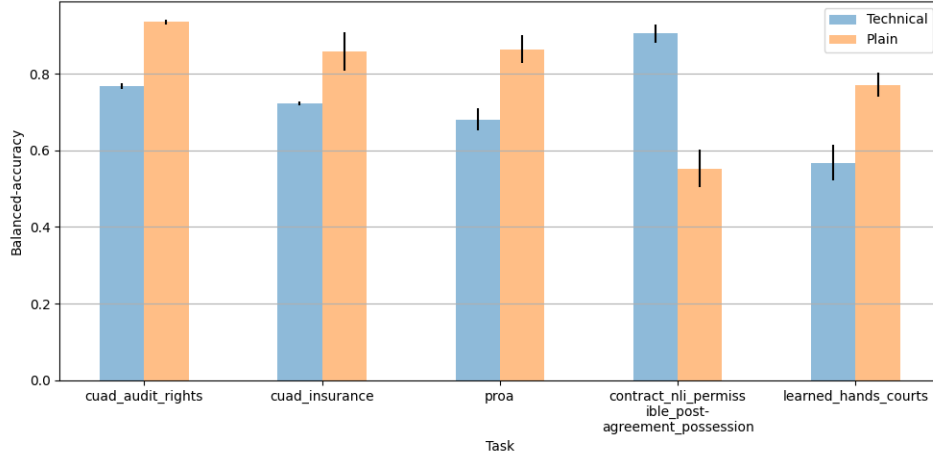


Figure 2: We compare performance of prompts which describe the task in plain language to prompts which describe the task in technical legal language (for GPT-3.5). Error bars measure standard error, computed using a bootstrap with 1000 resamples.

We explore this question by comparing “plain language” and “technical language” prompts. For a subset of LEGALBENCH tasks, we have access to the formal language provided to law-trained annotators when creating task data. By comparing the performance of a prompt which uses this language—to one which uses a plain-language version—we can measure how the technicality of language affects results.

We conduct preliminary experiments on a sample of five LEGALBENCH tasks (Figure 2).¹² On four of the five tasks, we find that the plain-language prompt significantly outperforms the technical language prompt, by up to 21 points (balanced-accuracy). Interestingly, on `contract_nli_permis-ible_post-agreement_possession`, we find the opposite phenomenon holds: the plain language prompt is substantially *worse* than the technical prompt.

Sensitivity to in-context demonstrations Finally, we investigate the influence of the in-context demonstrations used in prompts. Prior work in general domain LLMs have observed that few-shot performance is highly sensitive to the choice of demonstrations [58, 125, 143]. We evaluate whether LLMs are similarly sensitive for legal tasks, focusing on a subset of 8 binary classification tasks. For each task we merge the train and evaluation split into a single dataset, and randomly sample four in-context samples to include in the prompt (two from each class), five different times. We evaluated GPT-3.5 and Incite-Instruct-7B with each of the five generated prompts, and plot the balanced-accuracy of each prompt in Figure 3.

¹²Prompts are made available in the LEGALBENCH repository.